HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented electrical distribution systems for commercial and residential buildings, ensuring compliance with industry standards and sustainability goals.
- Developed detailed lighting designs and security system specifications for various project sectors, including healthcare and education facilities.
- Collaborated with multi-disciplinary teams to integrate building services designs into overall project blueprints, focusing on energy efficiency.
- Translated client requirements into technical specifications for electrical building services, including power generation and distribution strategies.
- Analyzed building energy performance data to inform design decisions and recommend sustainable engineering solutions.
- Contributed to the development of parametric modelling techniques for building design, specifically exploring applications with **Python** for automated analysis.
- Managed and executed change control processes for system configurations and updates, coordinating user acceptance testing and post-go-live support across interconnected systems.

SKILLS

Engineering – Electrical Engineering, Building Services, Critical Systems Design, Sustainable Engineering, Power Distribution, Lighting Design, Security Systems

Software & Scripting – Python, C#, VB.NET, Shell Scripting

Design Tools – Grasshopper, Dynamo, AutoCAD, Revit

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Databases & Scripting – PostgreSQL, MySQL, Oracle, SQL, Python scripting

Methodologies – Agile, Scrum, Change Management, Incident Response

PROJECTS

Sustainable Building Electrical Design Automation

- Developed a **Python** script to automate the calculation of electrical load requirements for diverse building types, reducing manual analysis time by approximately 40%.
- Integrated the script with building information models (BIM) using a prototype plugin for **Dynamo**, enabling real-time feedback on design choices against energy efficiency targets.
- Designed and documented the architecture for a system to monitor and report on energy consumption patterns within commercial properties, mapping sensor data to actionable insights.

Critical Systems Power Infrastructure Analysis

- Analyzed power generation and distribution requirements for critical infrastructure projects, such as data centers, ensuring high availability and redundancy.
- Wrote **SQL** queries against a PostgreSQL database to extract and validate historical power usage data from multiple sites, identifying patterns for capacity planning.
- Created comprehensive technical documentation and presentation materials for stakeholders, outlining proposed electrical system upgrades and their sustainability benefits.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)